Huawei Access Node KOR-ACC-29 Standard Operating Procedure for QoS Congestion Event (Microbursts)

Node: KOR-ACC-29 | **Vendor:** Huawei | **Severity:** Critical | **Alarm Code:** ALARM_5058

Description

This Standard Operating Procedure (SOP) is designed to guide the resolution of Critical QoS Congestion Events on Huawei Access Node KOR-ACC-29 at the Koramangala Datacenter. The failure domain in question revolves around microbursts, interface tail drops, and QoS queue starvation, resulting from the node's inability to handle the traffic load within the predetermined Quality of Service (QoS) parameters. This issue is exacerbated by the vendor's CLI dialect and the specific role of the access node in the network topology. In Huawei's OceanStor environment, the access node acts as a gateway between the access network and the core network, responsible for forwarding packets and enforcing QoS policies. The congestion event occurs when the node's QoS queues become overwhelmed, leading to packet drops and subsequent network instability. This SOP aims to identify the root cause of the issue, isolate the problem using diagnostic commands, and provide a step-by-step resolution process to restore network stability.

Pre-requisite Checks

- Verify that the Koramangala Datacenter is accessible and the node is powered on.

- Ensure that the remote hands team is dispatched to the datacenter to facilitate any necessary physical interventions.

- Check the status of redundant links to ensure that they are functioning properly.

- Verify that the node's software version is up-to-date and compatible with the network architecture.

Potential Root Causes

Cause	Probability
Misconfigured QoS policies	High
Hardware resource constraints (e.g., insufficient buffer size)	Medium
Software bugs or glitches affecting QoS management	Low

Diagnostic Commands & Expected Outputs

```
Command: display qos policy status, Expected Output: Policy name : qos-policy-1 Priority : 2 Bandwidth : 1000000 , Description: This command displays the current QoS policy settings, including priority, bandwidth, and other relevant parameters. This helps to identify potential misconfigurations or deviations from expected settings.
```

```
Command: display interface queue statistics, Expected Output:
Interface : 10GE1/0/0 Queue ID : 2 Packet drops : 1234 , Description:
This command provides statistics on packet drops, including the number
of dropped packets, which helps to determine if the issue is related
to interface tail drops or QoS queue starvation.

Command: display cpu utilization, Expected Output: CPU utilization :
90% , Description: This command displays the current CPU utilization,
which helps to determine if the issue is related to hardware resource
constraints.
```

Resolution Steps

Step 1: Step: Update QoS policies, Command: config qos policy update qos-policy-1 priority 1 bandwidth 5000000, Description: Update the QoS policy to increase priority and bandwidth to alleviate congestion on the access node.

Step 2: Step: Increase hardware resources (e.g., buffer size), Command: config interface buffer-size 10GE1/0/0 2048, Description: Increase the buffer size on the affected interface to reduce packet drops and alleviate congestion.

Step 3: Step: Perform a software upgrade, Command: upgrade software version 8.0.0, Description: Perform a software upgrade to the latest version to address potential software bugs or glitches affecting QoS management.

Post-Check Verification

```
Command: display qos policy status, Expected Output: Policy name :
qos-policy-1 Priority : 1 Bandwidth : 5000000 , Description: Verify
that the QoS policy has been updated successfully and the priority and
bandwidth have increased.

Command: display interface queue statistics, Expected Output:
Interface : 10GE1/0/0 Queue ID : 2 Packet drops : 0 , Description:
Verify that packet drops have decreased significantly, indicating that
the issue has been resolved.

Command: display cpu utilization, Expected Output: CPU utilization :
50% , Description: Verify that CPU utilization has decreased,
indicating that hardware resource constraints have been alleviated.
```

Roll-back Procedures

- Step: Rollback QoS policy update, Command: config qos policy rollback qos-policy-1, Description: Rollback the QoS policy update to its previous state in case the fix has not resolved the issue.

- Step: Rollback hardware resource changes, Command: config interface buffer-size 10GE1/0/0 1024, Description: Rollback the buffer size changes to their previous state in case the fix has not resolved the issue.

- Step: Rollback software upgrade, Command: upgrade software rollback version 7.0.0, Description: Rollback the software upgrade to the previous version in case the fix has not resolved the issue.